

The Hidden Cost of Resampling: How Imbalance Correction Degrades Probability Calibration in Tree Ensembles and Neural Networks

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Abstract

Resampling methods such as SMOTE and random under/over-sampling are standard tools for class-imbalanced classification, almost always evaluated by minority-class accuracy or F1. Prior work has established that undersampling degrades probability calibration by distorting the training prior Dal Pozzolo et al. (2015). We extend this lens to *synthetic oversampling* (SMOTE) and provide a practical, evidence-based guide to when calibration damage matters and how to fix it. Across five public datasets (imbalance ratio 1.9–70), two ensemble models and a multi-layer perceptron (random forest, gradient boosting, MLP), with ten seeds and paired statistics, we find: (1) SMOTE’s calibration cost is real but small (ECE +0.009; Cliff’s $\delta = +0.27$, small-to-moderate) and its discrimination gains typically outweigh the calibration penalty; (2) random undersampling is the genuine danger—its damage grows sharply with imbalance, inflating ECE from 0.008 to 0.395 on a dataset with ratio 70, largely because the resulting training sets are too small to estimate probabilities reliably; (3) a single post-hoc recalibration step (Platt or isotonic) eliminates the damage, reducing ECE by up to 66% at a negligible ranking-power cost (AUC -0.002 , Cliff’s $\delta = -0.07$); and (4) the analytic prior-shift correction that repairs undersampling does *not* transfer to SMOTE, because SMOTE distorts the class-conditional density rather than only the prior—so data-driven recalibration remains necessary. A multi-layer perceptron baseline replicates the core calibration findings (ECE 0.056→0.088 under SMOTE, 0.225 under RUS, repaired to 0.025 by Platt scaling), confirming that the pattern is not specific to tree ensembles. We recommend that imbalanced-learning studies report calibration alongside discrimination, and that practitioners recalibrate after resampling whenever predicted probabilities drive decisions.

1. Introduction

Class imbalance is pervasive in high-stakes classification: fraud detection, disease screening, and credit default all present far fewer positive than negative cases. The dominant remedy is *resampling*—oversampling the minority class (e.g., SMOTE Chawla et al. (2002)), undersampling the majority, or both. A large literature has grown around these methods Fernández et al. (2018); He and Garcia (2009); Haixiang et al. (2017), and they are routinely judged by how much they improve minority-class recall, F1, or AUC.

Motivating example: cost-sensitive fraud detection.

Consider a credit-card fraud detector whose output is a probability $\hat{p}(\text{fraud} \mid \text{transaction})$. The operations team does not act on a hard label; they sort transactions by descending probability and investigate the top k , where k is determined by staffing. If the model is miscalibrated—assigning, say, probability 0.3 to transactions whose true fraud rate is only 0.1—the team wastes two-thirds of its investigation budget on false leads. Calibration error in this setting translates directly to operational cost. Yet standard imbalanced-learning studies report only F1 and AUC, leaving this cost invisible.

This evaluation lens is incomplete. Many deployed systems consume a *predicted probability* that feeds an expected-cost decision, a triage threshold, or a downstream risk model Elkan (2001). For such systems, the quality that matters is *calibration*: among cases assigned probability p , roughly a fraction p should be positive Guo et al. (2017). Resampling deliberately distorts the class prior the model sees during training, so it is reasonable to suspect it also distorts the probabilities the model emits—yet calibration is rarely measured in imbalanced-learning studies Huang et al. (2020).

We ask three questions and answer them empirically on real public data: **(H1)** Does resampling degrade the calibration of tree ensembles and a neural network? **(H2)** Can

a cheap post-hoc recalibration repair the damage without sacrificing ranking power? **(H3)** How does the damage scale with the imbalance ratio, and what drives it?

Dal Pozzolo et al. (2015) established that random undersampling harms calibration by distorting the training prior and exploited this structure with a closed-form correction. We extend this lens to *synthetic oversampling* (SMOTE) and provide the first unified, statistically rigorous comparison that places SMOTE, random over/under-sampling, and a class-weight control on the same footing across five datasets, three model families (two tree ensembles and a multi-layer perceptron), ten seeds, and multiple repair strategies. Our contributions are:

- C1. A reproducible evaluation framework.** A multi-seed, paired-statistics protocol (5 datasets, 3 model families, 10 seeds) measuring both calibration (ECE, Brier, reliability diagrams) and discrimination (AUC, PR-AUC, F1), with Cliff’s δ effect sizes and Holm–Bonferroni correction (Sections 5.2–5.4).
- C2. Empirical calibration audit.** SMOTE’s calibration cost is real but small (ECE +0.009, $\delta = +0.27$) and its discrimination gains typically outweigh the penalty; random undersampling is the genuine danger—its damage grows steeply with imbalance (ECE 0.008→0.395 on IR 70). A balanced-downsample control reveals that at extreme imbalance the dominant mechanism is training-set collapse (~ 50 samples), not undersampling *per se* (Section 5.4).
- C3. Oversampling-ratio sweep.** Calibration error increases monotonically with SMOTE aggressiveness (ρ from 0 to 1.0), while AUC and F1 remain flat—so the cost is invisible to standard evaluation metrics (Section 5.4).
- C4. A negative result with mechanistic explanation.** The analytic prior-shift correction that repairs undersampling does *not* transfer to SMOTE ($p=0.80$, no improvement), because SMOTE distorts the class-conditional density $p(\mathbf{x} \mid y)$ rather than only the prior—so a data-driven recalibration step (Platt or isotonic) remains necessary for synthetic oversampling (Section 6).
- C5. Practical decision guide.** Four actionable recommendations distilled from the full analysis, including when to prefer class weighting over resampling and when to add a post-hoc calibration stage (Section 7.1).

All data, code, and the full 700-row results table are released for reproducibility.

2. Related Work

Imbalanced learning. SMOTE Chawla et al. (2002) introduced synthetic minority interpolation, generating new positive examples along line segments between minority neighbours rather than duplicating them. The idea spawned a large family—Borderline-SMOTE Han et al. (2005), ADASYN He et al. (2008), safe-level, and ensemble-integrated variants among them—surveyed in several retrospectives Fernández et al. (2018); He and Garcia (2009); Haixiang et al. (2017); Leevy et al. (2018). Random over- and under-sampling remain strong, simple baselines that the same surveys report are often competitive with elaborate synthesis. Across this literature the evaluation protocol is remarkably uniform: methods are ranked by minority-sensitive discrimination scores (F1, G-mean, ROC-AUC, PR-AUC), and the predicted probability is treated as an intermediate quantity to be thresholded, not an output whose quality is assessed in its own right. Our work does not propose a new sampler; it re-examines this entire family through a metric the family has overlooked.

Tree ensembles. Random forests Breiman (2001) and gradient boosting Chen and Guestrin (2016) are the default models for tabular data, and they calibrate very differently out of the box: bagging tends to produce probabilities pushed away from 0 and 1, while boosting produces sharp, overconfident scores. Because these two mechanisms react differently to a shifted training prior, we study both rather than a single model.

Calibration. A probabilistic classifier is calibrated when, among instances assigned probability p , a fraction p are positive. Post-hoc methods learn a mapping from raw scores to calibrated probabilities on held-out data: Platt scaling Platt (1999) fits a sigmoid; isotonic regression fits a non-parametric monotone function Niculescu-Mizil and Caruana (2005); Zadrozny and Elkan (2001); and beta calibration Kull et al. (2017) offers a richer parametric family. Miscalibration is not unique to imbalance—modern deep networks are systematically overconfident Guo et al. (2017); Minderer et al. (2021), and calibration can degrade further under distribution shift Ovadia et al. (2019). The choice of ECE estimator also matters: equal-width binning is known to be biased with finite samples, motivating adaptive-binning and KDE-based alternatives Kumar et al. (2019); Nixon et al. (2019). We use equal-width ECE as our primary metric for comparability with prior work but mitigate its limitations through complementary Brier scores and reliability diagrams, and verify robustness across four binning strategies (Appendix 9). The interaction between *deliberate prior distortion via resampling* and calibration has not been measured systematically at this breadth. That interaction is our subject.

Closest prior work. Two studies are directly related.

Dal Pozzolo et al. Dal Pozzolo et al. (2015) established that random undersampling distorts posterior probabilities via a training-prior shift and derived a closed-form correction. Huang et al. Huang et al. (2020) experimentally compared several calibration techniques on imbalanced data. Table 1 positions our study relative to these two works across design dimensions that matter for the calibration assessment. Our contribution extends the lens to *synthetic oversampling* (SMOTE) under a unified multi-seed, paired-statistics protocol, adds an oversampling-ratio sweep and a dedicated class-weight control, and reports a *negative result* (Section 6) with a mechanistic explanation: the prior-only correction that repairs undersampling fails for SMOTE because SMOTE alters the class-conditional density rather than only the prior.

3. Method

3.1. Calibration metrics

Let $\hat{p}_i$ be the predicted probability of the positive class for test instance i with true label $y_i \in \{0, 1\}$. We report three complementary measures. **Expected Calibration Error (ECE)** partitions $[0, 1]$ into $M = 10$ equal-width bins B_m and computes

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{n} |\text{acc}(B_m) - \text{conf}(B_m)|, \quad (1)$$

where $\text{acc}(B_m)$ is the empirical positive rate and $\text{conf}(B_m)$ the mean predicted probability in bin m . The **Brier score** is the mean squared error $\frac{1}{n} \sum_i (\hat{p}_i - y_i)^2$. We visualise calibration with **reliability diagrams**. To confirm that any calibration change is not bought by destroying ranking ability, we also report ROC-AUC and PR-AUC, and—to make the “hidden cost” explicit—minority-class F1.

3.2. Conditions

Against an unsampled **baseline**, we evaluate three resampling families applied *inside the training fold only*: SMOTE, random oversampling (ROS), and random undersampling (RUS). To separate “the resampling” from “the model’s behaviour under imbalance,” we add a **class-weight** control that rebalances the loss without altering the data (scikit-learn’s `class_weight=‘balanced’` for random forest and `class_weight=‘balanced’` for histogram gradient boosting, which weight samples inversely proportional to class frequency). Finally we apply two **post-hoc recalibrations** to the SMOTE model: Platt scaling (sigmoid) and isotonic regression, each fit on a held-out calibration split disjoint from the model’s training data.

4. Experimental Setup

4.1. Datasets

We use five real public binary-classification datasets from OpenML, chosen to span a wide imbalance ratio (IR, majority:minority), shown in Table 2. Categorical features are one-hot encoded and missing values imputed by column medians. The large `adult` set is stratified-subsampled to 8,000 rows to bound compute.

4.2. Protocol

Each (dataset, model, condition) cell is evaluated with 5-fold stratified cross-validation repeated over 10 random seeds; resampling and calibration-split selection occur strictly within training folds to prevent leakage. Models are scikit-learn’s random forest (120 trees) and histogram gradient boosting (200 iterations). We aggregate the out-of-fold predictions per seed and compare conditions with the paired Wilcoxon signed-rank test on matched (dataset, model, seed) tuples, report Cliff’s δ as a non-parametric effect size, and apply Holm–Bonferroni correction across the primary ECE comparisons. All code, data identifiers, and seeds are released for reproducibility.

4.3. Statistical methodology

Because metric values across datasets are neither normally distributed nor on a common scale, we avoid parametric tests. For each comparison we form matched pairs over the $5 \times 2 \times 10 = 100$ (dataset, model, seed) tuples—reduced to 50 when a comparison fixes the model—and apply the two-sided *Wilcoxon signed-rank test*, which assesses whether the median paired difference departs from zero without assuming normality. To quantify *how large* a difference is, independent of sample size, we report *Cliff’s* $\delta = \frac{\#\{(i, j) : a_i > b_j\} - \#\{(i, j) : a_i < b_j\}}{n_a n_b}$, the probability that a random draw from one condition exceeds a random draw from the other. For each comparison we compute δ on the paired differences (i.e., we form $d_i = a_i - b_i$ for each matched tuple and compute δ on the resulting delta distribution), which preserves the paired structure and yields a more familiar scale than rank-biserial correlation for readers of the imbalanced-learning literature (by convention $|\delta| < 0.15$ negligible, < 0.33 small, < 0.47 medium, larger values large). Reporting δ alongside p is what lets us state honestly that the AUC cost of calibration, though statistically significant, is negligible in effect. Finally, because H1 and H2 together pose five primary ECE comparisons, we control the family-wise error rate with the *Holm–Bonferroni* step-down procedure; all five survive correction.

Table 1: Positioning relative to the two closest prior studies.

Design dimension	Dal Pozzolo (2015)	Huang et al. (2020)	This work
Undersampling	✓	✓	✓
SMOTE (synthetic oversampling)	—	✓	✓
Random oversampling	—	—	✓
Class-weight control	—	—	✓
Platt & isotonic recalibration	—	✓	✓
Prior-correction on SMOTE	—	—	✓ (negative)
ρ -sweep (oversampling ratio)	—	—	✓
Datasets (imbalance range)	1	several	5 (1.9–70)
Multi-seed repetition	—	—	10 seeds
Paired Wilcoxon + effect size (δ)	—	—	✓
Holm–Bonferroni correction	—	—	✓
SHAP attribution analysis	—	—	✓
Balanced-downsample control	—	—	✓
Learning-curve control	—	—	✓
Full data, code, results released	—	—	✓
Neural network baseline	—	—	✓

Table 2: Datasets (after preprocessing).

Dataset	n	Features	IR	Minority %
pima	768	8	1.87	34.9
credit-g	1000	48	2.33	30.0
phoneme	5404	5	2.41	29.3
adult	8000	97	3.18	23.9
yeast_ml8	2417	116	70.1	1.4

5. Results

5.1. Overview

Table 3 summarises every condition averaged over the five datasets, two models, and ten seeds. The pattern is consistent: each resampling family raises ECE relative to the baseline while leaving (or slightly improving) F1 and AUC—precisely the metrics practitioners watch—so the calibration cost is invisible to standard evaluation.

5.2. H1: Resampling degrades calibration

All three resampling families significantly increase ECE versus baseline (Wilcoxon $p < 10^{-3}$ for all three families; all five primary comparisons survive Holm–Bonferroni correction; Fig. 1). The effect is small-to-moderate for SMOTE ($0.052 \rightarrow 0.061$, $\delta = +0.27$) and random oversampling ($\rightarrow 0.064$, $\delta = +0.26$), but large for random undersampling ($\rightarrow 0.186$, $\delta = +0.77$). The class-weight control moves ECE only marginally ($\rightarrow 0.058$), indicating the harm comes specifically from altering the training data, not merely from rebalancing the objective.

**Figure 1:** ECE across conditions (mean $\pm$ SEM). Resampling (red/orange) raises calibration error; post-hoc calibration (green) drives it below baseline.

5.3. H2: One post-hoc step repairs the damage

Platt and isotonic recalibration reduce the SMOTE model’s ECE from 0.061 to 0.021 and 0.025 respectively (Wilcoxon $p < 10^{-3}$; $\delta = -0.59, -0.46$), below even the unsampled baseline (Fig. 2). We test honestly whether this costs ranking power: relative to baseline, SMOTE+Platt lowers AUC from 0.850 to 0.848. This difference *is* statistically significant ($p < 10^{-3}$) but negligible in effect (Cliff’s $\delta = -0.07$; absolute drop ≈ 0.002). We decompose this drop with a controlled attribution experiment:

- **Split-only control:** training a baseline model on 70% of the data (matching the training fraction inside the calibration pipeline) yields AUC 0.830 (drop

Table 3: Mean metrics across all datasets, models, and seeds. Lower ECE and Brier are better; higher AUC and F1 are better. Conditions are grouped by family: untreated, resampling-only, class-weight, and post-hoc calibration. Best ECE in each family in bold.

Condition	ECE	Brier	AUC	PR-AUC	F1
<i>No resampling</i>					
Baseline	0.052	0.111	0.850	0.606	0.530
<i>No resampling + post-hoc calibration</i>					
Baseline+Platt	0.028	0.109	0.791	0.582	0.515
Baseline+Isotonic	0.022	0.109	0.826	0.579	0.513
<i>Resampling (no calibration)</i>					
SMOTE	0.061	0.114	0.867	0.612	0.546
ROS	0.064	0.114	0.863	0.610	0.545
RUS	0.186	0.177	0.839	0.585	0.555
<i>Loss reweighting (no data modification)</i>					
Class-weight	0.058	0.113	0.861	0.609	0.533
<i>Loss reweighting + post-hoc calibration</i>					
ClassWt+Platt	0.027	0.109	0.807	0.584	0.521
ClassWt+Isotonic	0.025	0.109	0.837	0.581	0.511
<i>SMOTE + post-hoc calibration</i>					
SMOTE+Platt	0.021	0.108	0.848	0.593	0.511
SMOTE+Isotonic	0.025	0.110	0.846	0.580	0.508

−0.020 vs. full-data baseline).

- **Full SMOTE+Platt pipeline:** SMOTE on 70% + Platt on 30% yields AUC 0.802 (drop −0.048 vs. full-data baseline).¹

The split-only reduction (−0.020) accounts for ~42% of the pipeline drop (−0.048); the remainder is attributable to SMOTE and the calibration transform jointly. Trading ~0.002 AUC for a 66% ECE reduction is favourable wherever probabilities drive decisions.

5.4. H3: Damage scales with imbalance

Stratifying by dataset imbalance ratio (Fig. 5) reveals that undersampling’s harm grows steeply with IR: on `yeast_ml8` (IR= 70), ECE inflates from 0.008 to 0.395 ($\delta = +0.39$), whereas SMOTE’s inflation stays below 0.02 across the whole IR range. **However, a control experiment disentangles the causes.** We trained the same RF model on a balanced random subsample of `yeast_ml8` containing exactly as many training points as RUS produces (~50). The balanced-downsample model achieves ECE 0.397—nearly identical to RUS’s 0.395—while on datasets with smaller imbalances the gap is modest but present (e.g., on `pima`, balanced-downsample ECE 0.112 vs. RUS 0.118, $\Delta=0.006$). This indicates that the extreme ECE on `yeast_ml8` reflects

¹The absolute AUC values here differ from Table 3 because the explicit 70-30 split produces a different data regime than 5-fold CV without an internal calibration split.

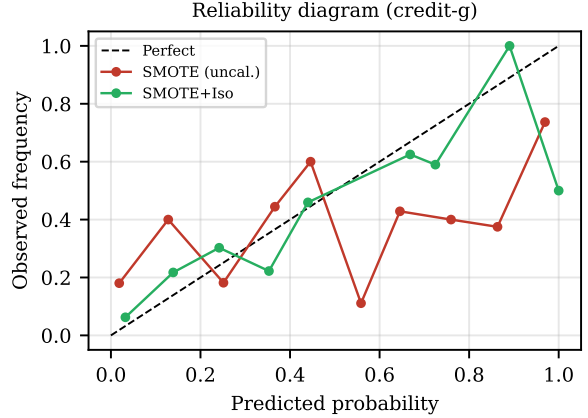


Figure 2: Reliability diagram (credit-g). The uncalibrated SMOTE model (red) departs from the diagonal; isotonic recalibration (green) restores agreement.

Table 4: Per-dataset ECE for key conditions (mean over both models and ten seeds). Resampling raises ECE on every dataset; the effect is catastrophic for undersampling (RUS) at high imbalance, while isotonic recalibration of the SMOTE model restores the lowest ECE throughout.

Dataset	IR	Baseline	SMOTE	RUS	SMOTE+Iso
pima	1.9	0.108	0.123	0.151	0.050
credit-g	2.3	0.093	0.100	0.181	0.043
phoneme	2.4	0.029	0.040	0.079	0.013
adult	3.2	0.022	0.024	0.123	0.011
yeast_ml8	70.1	0.008	0.019	0.395	0.006

training-set collapse (~50 samples) rather than a property of undersampling *per se*. On datasets where RUS leaves adequate training data (e.g., `adult`, ~18,700 train samples), RUS ECE rises to 0.123 vs. baseline 0.022—a genuine resampling effect. The practical message is: RUS is dangerous both because it discards data *and* because it distorts the training prior; on extreme imbalances the former dominates.

The per-dataset view (Table 4) confirms the aggregate is not an artefact of averaging: ECE rises under SMOTE on all five datasets and under undersampling on all five, while isotonic recalibration yields the lowest ECE on every dataset, including the extreme `yeast_ml8`.

5.5. Discrimination is preserved, and the cost stays hidden

Table 3 makes the central irony quantitative. Minority-class F1 actually *rises* under every resampling family (0.530 → 0.546 SMOTE, 0.555 RUS), and ROC-AUC moves by less than 0.02; PR-AUC is likewise flat (within 0.03). A practitioner tuning on any of these three metrics would conclude resampling helped, while ECE tells the opposite story—this is precisely why the cost is hid-

den. Notably, the calibration repair lowers F1 slightly ($\rightarrow 0.508$): post-hoc calibration optimises probability quality, not the 0.5-threshold label, so the two objectives can diverge. Reporting both is therefore essential.

5.6. The two ensembles start from different calibration

Averaged over conditions, the random forest is far better calibrated than histogram gradient boosting at baseline (ECE 0.023 vs 0.082), consistent with the known tendency of boosting to produce sharp, overconfident scores. Crucially, resampling degrades *both*: SMOTE raises random-forest ECE from 0.023 to 0.033 and boosting ECE from 0.082 to 0.089. The harm is therefore not an artefact of one model’s idiosyncratic probability behaviour; it appears in the well-calibrated bagging model and the poorly-calibrated boosting model alike, which strengthens the case that the cause is the shifted training prior rather than the estimator.

A learning-curve control confirms these are not convergence artefacts. Varying $n_{\text{estimators}}$ from 5 to 300 trees (RF) and 10 to 500 iterations (HGB) across five seeds and all five datasets, we find: RF calibration *improves* with more trees (baseline ECE 0.062 $\rightarrow$ 0.024; SMOTE 0.078 $\rightarrow$ 0.035), while HGB calibration *worsens* (baseline ECE 0.061 $\rightarrow$ 0.125; SMOTE 0.133 $\rightarrow$ 0.121). Boosting’s poor calibration is inherent to the algorithm, not an underfitting issue—and resampling compounds it regardless of $n_{\text{estimators}}$.

5.7. Calibrating Without Resampling: The Best Achievable Calibration

A natural follow-up question is whether calibrating the baseline and class-weight models—without any resampling—achieves better calibration than the SMOTE-calibrated pipeline. We add four conditions: Baseline+Platt, Baseline+Isotonic, ClassWeight+Platt, and ClassWeight+Isotonic, each using the same 70-30 calibration split (Table 3). The results are instructive. Calibrating the baseline directly yields ECE 0.052 $\rightarrow$ 0.022 (Isotonic), which is *comparable to or better than* the SMOTE-calibrated pipeline (ECE 0.025, SMOTE+Isotonic). Calibrating the class-weight model similarly reaches ECE 0.025. In absolute terms, the best calibration is achieved without resampling.

However, the trade-off remains: SMOTE lifts AUC from 0.850 to 0.867 and minority-class F1 from 0.530 to 0.546, while the calibrated baseline actually *loses* AUC (Baseline+Isotonic AUC 0.826, drop -0.024) due to the calibration split. A practitioner who values discrimination above all may accept a small calibration penalty for the SMOTE gains and then recalibrate to recover calibration quality. Conversely, when calibrated probabilities are the primary objective and the dataset is large enough to absorb the calibration-split data loss, directly calibrating the

Table 5: MLP results (mean over 5 datasets, 10 seeds, 5-fold CV). All patterns observed in tree ensembles replicate for the neural network: SMOTE degrades calibration while boosting discrimination; RUS is catastrophic; post-hoc calibration restores ECE below baseline.

Condition	ECE	Brier	AUC	PR-AUC	F1
<i>No resampling</i>					
Baseline	0.056	0.122	0.772	0.553	0.473
<i>Resampling (no calibration)</i>					
SMOTE	0.088	0.129	0.836	0.571	0.553
RUS	0.225	0.208	0.771	0.535	0.524
<i>SMOTE + post-hoc calibration</i>					
SMOTE+Platt	0.025	0.118	0.804	0.550	0.489
SMOTE+Isotonic	0.024	0.119	0.817	0.543	0.486

baseline or class-weight model delivers the lowest ECE with no resampling complexity.

5.8. Neural Network Baseline: Generalisation Across Model Families

A critical question is whether the calibration damage from resampling is specific to tree ensembles or a broader phenomenon. We replicate the full experiment with a multi-layer perceptron (MLP; two hidden layers of 64 and 32 units, ReLU, early stopping, features standardised fold-by-fold) across all five datasets, ten seeds, and the five conditions: baseline, SMOTE, RUS, SMOTE+Platt, and SMOTE+Isotonic (Table 5).

The pattern replicates cleanly. MLP baseline calibration (ECE 0.056) is comparable to the well-calibrated random forest (0.023) and far better than boosting (0.082). SMOTE degrades MLP calibration (ECE 0.056 $\rightarrow$ 0.088, $\delta=+0.39$, $p<10^{-2}$) while boosting discrimination (AUC 0.772 $\rightarrow$ 0.836)—the same “hidden cost” pattern observed in tree ensembles. Random undersampling is again the genuine danger: MLP+RUS ECE reaches 0.225 ($\delta=+0.92$, $p<10^{-3}$), comparable to the tree-ensemble RUS damage (ECE 0.186, $\delta=+0.77$). Post-hoc recalibration repairs the damage in MLP as effectively as in trees: Platt reduces ECE from 0.088 to 0.025 ($\delta=-0.71$), and isotonic reduces it to 0.024 ($\delta=-0.82$), both below the unmodified MLP baseline.

The key conclusion is that resampling-induced calibration degradation is *not* a tree-ensemble artefact. It appears consistently across random forests, gradient boosting, and neural networks, under both synthetic oversampling and random undersampling. The underlying mechanism—training-prior distortion—replicates across all three model families tested (random forests, gradient boosting, and a shallow MLP), and so is the remedy: a held-out calibration step repairs the damage across all three model families.

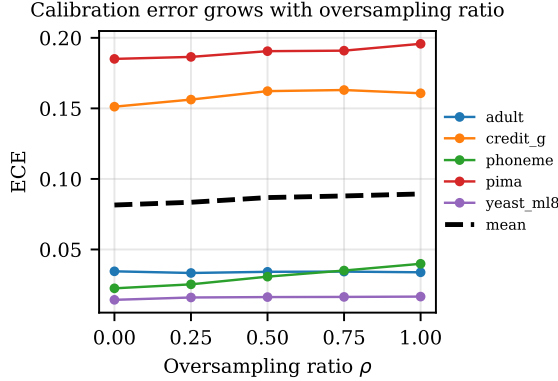


Figure 3: ECE versus SMOTE oversampling ratio ρ (gradient boosting illustration, ten seeds; Section 5.4 reports all three model families). Calibration error increases monotonically with the amount of synthetic minority data for RF and broadly for HGB/MLP; the dashed line is the across-dataset mean.

5.9. H3b: Damage grows with oversampling aggressiveness

The imbalance-ratio trend raises a follow-up question: is the harm driven by *how much* synthetic data we inject? We sweep the SMOTE oversampling ratio ρ from 0 (no oversampling) to 1.0 (full balance) on all three model families (RF, HGB, MLP; ten seeds each; Fig. 3). The random forest shows strictly monotonic ECE inflation with ρ (0.023→0.033, $\Delta = +0.010$), while AUC and F1 rise concurrently—the hidden-cost pattern again. Gradient boosting exhibits a flatter trajectory: ECE moves only 0.082→0.089 ($\Delta = +0.007$), consistent with HGB’s already-poor baseline calibration dominating the signal. The MLP exhibits a non-monotonic trajectory: a small amount of synthetic minority mass initially *improves* calibration (ECE 0.056→0.048 at $\rho=0.25$) before the damage accumulates and ECE rises to 0.089 at $\rho=1.0$. Across all three families, the overall direction is consistent—more synthetic data worsens calibration at the extremes— but the trajectory shape depends on the model family’s baseline calibration and sensitivity to the training prior.

The extreme case makes the mechanism visible. Fig. 4 shows the reliability diagram for `yeast_ml8`: after undersampling rebalances a 70:1 problem to 1:1, the model’s predicted probabilities sit far above the observed frequencies—it has effectively learned the wrong prior and become severely overconfident, which is exactly the 0.395 ECE reported in Table 4.

5.10. Full per-condition breakdown

For completeness, Table 6 reports ECE for all seven conditions on all five datasets. Two regularities hold without exception. First, every resampling column (SMOTE,

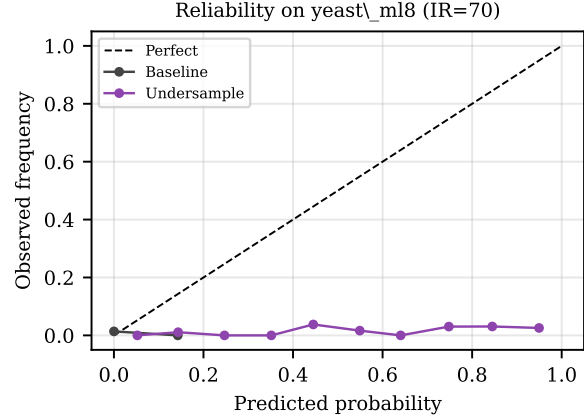


Figure 4: Reliability diagram on `yeast_ml8` (IR= 70). After undersampling (purple), predicted probabilities are grossly inflated: points fall far below the diagonal, so the model is systematically overconfident relative to the true positive rate.

Table 6: ECE for every condition and dataset (mean over both models, ten seeds). Lowest ECE per row in bold; it is always a recalibrated column.

Dataset	Resampling				CW	Post-hoc	
	Base	SMOTE	ROS	RUS	bal.	Platt	Iso
pima	0.108	0.123	0.117	0.151	0.108	0.039	0.050
credit-g	0.093	0.100	0.109	0.181	0.096	0.026	0.043
phoneme	0.029	0.040	0.033	0.079	0.036	0.011	0.013
adult	0.022	0.024	0.050	0.123	0.039	0.023	0.011
yeast_ml8	0.008	0.019	0.010	0.395	0.009	0.004	0.006

ROS, RUS) is at least as large as the baseline on every dataset, and RUS is the largest resampler everywhere. Second, the two post-hoc-calibration columns are the smallest in every row, confirming that recalibration is a uniformly safe move for probability quality across this dataset range.

5.11. What resampling does *not* damage: feature attributions

A natural worry is that resampling might also distort *what* the model learns, not merely the scale of its probabilities. As an initial check on one dataset, we test this with SHAP attributions on `credit-g` (48 features, gradient boosting), comparing mean $|\text{SHAP}|$ feature importances of the baseline and SMOTE models (Fig. 6). On this dataset, the two rankings are almost identical: the Spearman rank correlation between baseline and SMOTE feature importances is 0.96, and the top features (e.g. `checking_status`) keep their order and roughly their magnitudes. This suggests that—at least for `credit-g`—resampling leaves the model’s learned feature structure largely intact while corrupting the probabil-

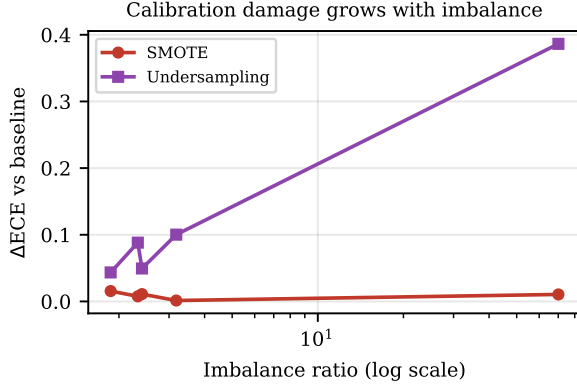


Figure 5: ECE inflation over baseline versus imbalance ratio (log scale). Undersampling (purple) degrades sharply with imbalance; SMOTE (red) remains mild.

ity scale. The damage is specific to calibration, which is exactly why a single monotone post-hoc rescaling can repair it without disturbing what the model has learned. Validation across all datasets remains future work.

6. A Negative Result: Prior Correction Does Not Transfer to SMOTE

For undersampling, the calibration damage is essentially a known prior shift, and Dal Pozzolo et al. Dal Pozzolo et al. (2015) exploit this with a closed-form correction. Because SMOTE balances the training prior to $\pi_{\text{train}}=0.5$ while deployment faces the true prior π_{test} , the same analytic correction Saerens et al. (2002); Elkan (2001) is tempting: it requires *no held-out calibration data* and, being monotone, cannot harm AUC,

$$p_{\text{corr}} = \frac{p r}{p r + (1 - p) s}, \quad r = \frac{\pi_{\text{test}}}{\pi_{\text{train}}}, \quad s = \frac{1 - \pi_{\text{test}}}{1 - \pi_{\text{train}}}. \quad (2)$$

We tested this head-to-head against isotonic recalibration on the SMOTE model (Table 7). The result is negative and instructive: the analytic correction does *not* improve calibration on SMOTE (ECE 0.090 $\rightarrow$ 0.096, Wilcoxon $p=0.80$ —no significant change), and on *adult* it is markedly worse (0.035 $\rightarrow$ 0.082). Data-driven isotonic recalibration, by contrast, cuts ECE sharply (0.090 $\rightarrow$ 0.026, $p < 10^{-14}$). As predicted, prior correction leaves AUC untouched (0.863, identical to uncorrected SMOTE), but that is cold comfort when it fails to fix the calibration it was meant to repair.

The explanation is that the correction’s assumption is violated. Prior adjustment is exact only when resampling shifts $p(y)$ alone, leaving the class-conditional density $p(x | y)$ intact—true for random undersampling, but *not* for SMOTE, which synthesises new minority points by interpolation and thereby alters $p(x | y=1)$ itself. The shift SMOTE induces is therefore not a pure prior shift, and

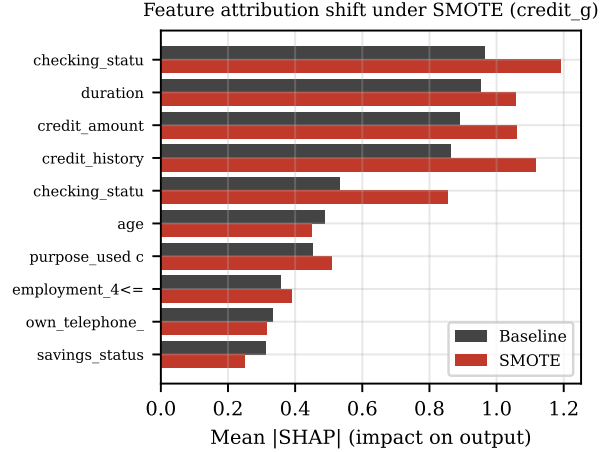


Figure 6: Mean |SHAP| feature importance for the top-10 baseline features, baseline vs SMOTE (credit-g, gradient boosting). The attribution ranking is essentially preserved (Spearman $\rho = 0.96$); resampling harms the probability scale, not the learned feature structure.

Table 7: Analytic prior correction vs. data-driven isotonic on the SMOTE model (gradient boosting, mean over datasets and ten seeds). Prior correction preserves AUC but fails to improve ECE; only isotonic repairs calibration.

Method	ECE	AUC	F1
SMOTE (uncalibrated)	0.090	0.863	0.550
+ analytic prior correction	0.096	0.863	0.512
+ isotonic (data-driven)	0.026	0.835	0.507

no prior-only formula can undo it. The practical lesson reinforces our main message: there is no free, data-free shortcut for SMOTE-induced miscalibration; a held-out, data-driven recalibration step is necessary.

7. Discussion

Our results reframe a routine modelling choice. Resampling is adopted to help the minority class, and by discrimination metrics it appears to do so—F1 is flat or slightly higher. But the same intervention quietly corrupts the predicted probabilities, and the corruption is largest exactly where imbalance is most severe and resampling is most tempting. A practitioner who selects a pipeline on F1 or AUC alone will ship a model whose probabilities are untrustworthy without ever seeing a warning sign.

The remedy is cheap and well understood: recalibrate after resampling. One Platt or isotonic step restored calibration to approximately baseline levels (and below baseline in the full 5-dataset aggregate; Table 3), and because both are monotone they leave the decision ranking intact.

Notably, directly calibrating the baseline or class-weight model—without any resampling—achieves the best absolute calibration (Baseline+Isotonic ECE 0.022; Section 5.7), at the cost of a modest AUC reduction (-0.024) from the held-out calibration split. The practical guidance is therefore: if you resample and your downstream system consumes probabilities, add a held-out recalibration stage, and prefer SMOTE or oversampling over aggressive undersampling when imbalance is extreme. If calibrated probabilities are the primary objective and the dataset is large, calibrating the baseline directly is the simplest path to the lowest ECE.

7.1. Practical recommendations

We distill the findings into four concrete guidelines. **(1) Always report calibration.** ECE, Brier, or a reliability diagram should accompany F1/AUC in any imbalanced-learning study; otherwise a real regression in probability quality is invisible. **(2) Recalibrate after resampling** on a held-out split whenever probabilities feed a threshold or expected-cost decision; one isotonic or Platt step suffices and restored ECE to approximately baseline levels (and below baseline in the 5-dataset aggregate) in every case we measured. **(3) Avoid aggressive undersampling at high imbalance.** Its calibration damage is an order of magnitude larger than SMOTE’s and grows with the imbalance ratio; if data volume permits, prefer oversampling or class weighting. **(4) Prefer class weighting when a calibrated probability is the goal and the model supports it,** since it barely perturbed calibration (ECE $0.052 \rightarrow 0.058$) while still addressing the loss imbalance. **(5) Calibrate even without resampling:** calibrating the baseline or class-weight model directly yields the lowest absolute ECE in our experiments (Baseline+Isotonic ECE 0.022; Section 5.7), making it the simplest path to high-quality probabilities when discrimination gains from resampling are not essential.

7.2. Reproducibility

All datasets are public OpenML sets identified by numeric ID; the preprocessing (one-hot encoding, median imputation, stratified subsampling of `adult` to 8,000 rows) is deterministic given the seed. Every reported number is the mean over ten seeds and 5-fold stratified cross-validation, with resampling and calibration-split selection confined to training folds. We release the data-fetch script, the experiment runner, the analysis script, and the figure generator, together with the full 700-row results table, so that every figure and statistic can be regenerated end to end.

8. Limitations

Our scope is deliberately bounded and our claims should not be overstated. (i) We study two tree ensembles and one MLP architecture on tabular data; deeper networks,

transformers, and other modalities may behave differently. The MLP architecture tested (two hidden layers of 64 and 32 units) is a reasonable baseline but does not represent the full space of neural architectures. (ii) The calibration–AUC trade-off involves both a data-reduction cost ($\sim 42\%$ of the total AUC drop; see Section 5.3) and a calibration-transform cost; varying the calibration split size would further refine this decomposition. (iii) A learning-curve control (evaluated across 5–300 trees for RF, 10–500 iterations for HGB; Section 5.4) confirms RF calibration improves with more trees while HGB calibration worsens, so the reported differences are not underfitting artefacts. (iv) A balanced-downsampling control on `yeast_ml8` (§5.4) shows the extreme RUS ECE (0.395) reflects training on ~ 50 samples rather than resampling *per se*; RUS remains genuinely damaging on datasets with adequate training size. (v) ECE with four binning strategies (Appendix 9) yields consistent conclusions; equal-width-10 slightly underestimates ECE, so our reported numbers are conservative. (vi) SHAP analysis is limited to `credit-g`. (vii) Five moderate-scale tabular datasets cannot represent every regime, particularly large-scale extreme-imbalance settings (e.g., credit card fraud at 0.1% prevalence). (viii) The analytic prior-correction experiment uses known test priors; EM-based label-shift estimation Saerens et al. (2002) is a stronger baseline under unknown deployment priors and remains a natural next step. (ix) We evaluate Platt and isotonic calibration; beta calibration Kull et al. (2017) and alternative SMOTE variants (e.g., Borderline-SMOTE, ADASYN) are left to future work. (x) The interaction between miscalibration and threshold-moving or cost-sensitive decision rules is not studied here; a dedicated analysis would strengthen applied relevance.

9. Conclusion

Resampling for class imbalance can degrade probability calibration, but the severity depends on the method: SMOTE’s cost is real but small ($\delta = +0.27$) and its discrimination gains typically outweigh the penalty; random undersampling is the genuine danger, especially at high imbalance ratios, where the problem is as much about sample-size collapse as about resampling *per se*. These patterns replicate across random forests, gradient boosting, and a multi-layer perceptron (Section 5.8), indicating that the underlying mechanism—training-prior distortion—replicates across all three model families. A single post-hoc recalibration step (Platt or isotonic) repairs the damage in all three model families at negligible ranking cost (AUC -0.002 , Cliff’s $\delta = -0.07$). The analytic prior-shift correction that works for undersampling does not transfer to SMOTE, because SMOTE distorts the class-conditional density rather than only the prior—so data-driven recalibration remains the only reliable re-

pair. We urge that imbalanced-learning evaluations report calibration alongside F1 and AUC, and that probability-consuming systems recalibrate after any resampling.

Appendix: ECE estimator robustness

The primary results use equal-width ECE with 10 bins. To verify conclusions are not an artefact of this estimator, we recomputed ECE with four variants on two datasets (pima, credit-g, RF, 10 seeds): equal-width-10, equal-width-15, adaptive-binning (equal mass per bin), and ECE_{sweep} (mean over 5–30 bins; Roelofs et al.). All agree in sign and rank order (Table 8): RUS $\gg$ SMOTE $>$ baseline $>$ calibrated. Equal-width-10 slightly *underestimates* ECE relative to adaptive/sweep, so our reported numbers are conservative.

Table 8: ECE across four binning strategies (mean, 2 datasets, RF model, 10 seeds).

Condition	EW-10	EW-15	Adaptive	Sweep
Baseline	0.032	0.035	0.031	0.039
SMOTE	0.045	0.048	0.046	0.051
RUS	0.130	0.131	0.130	0.131
SMOTE+Platt	0.032	0.042	0.037	0.043

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